Week 10

Course Level Objectives:

- Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. (CLO 3)
- Explain the role of probability in statistics. (CLO 4)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe Type I and Type II errors in the context of a study. (CLO 8) **2.** Relate the significance level α to the probability of a Type I error. (CLO 4, CLO 8) **3.** Explain the trade-off between Type I and Type II errors and how sample size affects it. (CLO 8) **4.** Explain how multiple testing and publication bias produce false positives. (CLO 8) **5.** Distinguish statistical significance from practical significance. (CLO 8)	e-Text: 4.4 A Closer Look at Testing Interactive Lecture: 4.4 A Closer Look at Testing Slides: 4.4 A Closer Look at Testing (includes Khan Academy Type I/II video)	Participation: Errors in Testing – battery-life example (a)–(d): describe each error in context; courtroom analogy matching; vitamin E multiple-testing questions Your Turn: Errors and Multiple Tests (ambulance response times)	Online Homework: 4.4 A Closer Look at Testing (a) (due Sun Nov 1, 11:59 PM) Online Homework: 4.4 A Closer Look at Testing (b) (due Sun Nov 1, 11:59 PM) Take-Home Quiz 6 (due Mon Nov 2, 11:59 PM) – covers 4.4–4.5 (all objectives) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Respondus
6. Describe how to generate randomization samples for a given study design (coin flips, re-randomizing groups, shifting data then bootstrapping). (CLO 8) **7.** Identify the centers of the bootstrap and randomization distributions for given data. (CLO 5) **8.** Use a 95% confidence interval to reach a two-tailed test conclusion at α = 0.05. (CLO 6, CLO 8) **9.** Determine whether a confidence interval or a hypothesis test fits a research question. (CLO 6, CLO 8)	e-Text: 4.5 Making Connections Interactive Lecture: 4.5 Making Connections Slides: 4.5 Making Connections	Participation: Making Connections – body temperature: test 98.6 using the CI (98.05, 98.47); sleep study (28/50): how to generate randomization samples Your Turn: CI or Test? (smart-speaker ownership)	Online Homework: 4.5 Making Connections (due Sun Nov 1, 11:59 PM) Take-Home Quiz 6 (due Mon Nov 2, 11:59 PM) – covers 4.4–4.5 (all objectives) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Respondus
10. Explain when the Central Limit Theorem justifies replacing a randomization distribution with a normal distribution N(null value, SE). (CLO 5) **11.** Compute a standardized z statistic. (CLO 8) **12.** Determine a p-value as a right-tail, left-tail, or two-tail area under the standard normal curve. (CLO 3, CLO 8)	e-Text: 5.1 Hypothesis Tests Using Normal Distributions Interactive Lecture: 5.1 Hypothesis Tests Using Normal Distributions Slides: 5.1 Hypothesis Tests Using Normal Distributions StatKey – Theoretical Distributions (Normal)	Participation: Normal Distributions and z – malaria/mosquito example: compute z, shade the tail in StatKey, find the p-value Your Turn: z-Tests (right-, left-, and two-tailed)	Online Homework: 5.1 Hypothesis Tests Using Normal Distributions (due Sun Nov 1, 11:59 PM) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Respondus

Also this week: Test 2 – Fri Oct 30 – Sun Nov 1 (Blackboard, proctored, 2 hours). Covers Weeks 6–9.